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INSTITUTE OF TECHNOLOGY



**International Conference on
Advanced Computing Systems and
Intelligent Communication Engineering (ICACSICE 2026)**

27th -28th March 2026

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UNDER SPARSE OBSERVATIONS

in International Conference on Advanced Computing Systems and Intelligent Communication Engineering (ICACSICE 2026) held on 27th and 28th March 2026 at Sambhram Institute of Technology, Bengaluru.

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Machine Learning-Enhanced EKF for Vehicle Localization under Sparse Observations

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Keywords: Extended Kalman Filter(EKF), Feed Forward Neural Network (FFNN), State Estimation

Abstract: This paper presents a hybrid method that combines the EKF with a NN to estimate a vehicle's position when a specific sensor model is unavailable or difficult to define. The vehicle's motion is modeled using linear differential equations, while sensor measurements are assumed to be nonlinear and partially unknown. To approximate the unknown measurement function, a multilayer feedforward NN is trained using synthetic sensor data. This NN is then integrated into the EKF framework to handle nonlinearities and compensate for the lack of explicit sensor models. The entire system is developed and tested in MATLAB. Simulation results demonstrate that the proposed approach provides accurate and reliable position estimates, maintaining robustness even under conditions of missing or uncertain sensor information.

1 INTRODUCTION

Accurate vehicle localization is a critical requirement in many real-world applications such as autonomous driving, ADAS and mobile robotics. In practice scenarios, sensor measurements can often be unreliable, degraded, or partially unavailable due to environmental factors, sensor faults, or cost constraints. Traditional methods such as the EKF rely heavily on precise sensor models, which may not always be available or accurately defined. This limitation can significantly reduce the reliability of position estimates and compromise system performance (Yuhao, 2022; Chatterjee & Maitra, 2025; Zhou et al., 2017).

To address this challenge, learning-based methods particularly NN are increasingly being integrated into estimation frameworks to approximate unknown or complex measurement models. Among the various types of neural networks, FFNNs have been widely employed for modeling vehicle dynamics (Chen et al., 2019; Zhang et al., 2021). In these applications, the FFNN typically acts as a black-box model, learning a neural representation of the underlying dynamical system from input and output data without requiring explicit knowledge of the physical model (Singh et al., 2020).

By combining data-driven learning with model-based filtering, hybrid approaches offer improved adaptability and robustness in uncertain or dynamic environments. This makes the estimation problem not only more challenging but also more relevant for industry, where maintaining accurate localization despite sensor limitations is essential for safety, efficiency and real-time operation (Fu & Luo, 2021; Zhou et al., 2017; Lu et al., 2022).

Deep learning and artificial NN have shown significant promise in vehicle trajectory and state prediction tasks due to their ability to model complex, nonlinear dynamics (Sushmitha et al., 2019; Zhang et al., 2018; Tang et al., 2024). Techniques such as long short-term memory (LSTM), Transformer, and feedforward networks have been applied for predicting vehicle trajectories (Qie et al., 2023) as well as vehicle speed and lateral and yaw behaviors (Ali et al., 2022). These models enhance situational awareness in ADAS and autonomous systems (Goforth & Lucey, 2019), especially under uncertain conditions where traditional model-based approaches may struggle (Jeong et al., 2017).

This research introduces a hybrid method that integrates the EKF with a NN to enhance vehicle state estimation in scenarios where precise sensor models

are unavailable. By employing a multi-layer FFNN to approximate unknown nonlinear measurement functions (Zha et al., 2022), the proposed system effectively addresses the limitations of traditional EKF approaches. Simulation results demonstrate that this integration leads to more accurate and reliable position estimates, even under conditions involving incomplete or poorly defined sensor models. This approach holds significant potential for improving vehicle localization in real-world applications where sensor information is often uncertain or incomplete. The main contributions of this paper are as follows:

- A hybrid estimation framework is proposed that integrates a multi-layer feedforward neural network with the Extended Kalman Filter (EKF), enabling the approximation of unknown nonlinear measurement functions commonly encountered in vehicle localization tasks.
- The proposed approach enables robust state estimation even in the absence of explicit or accurate sensor models or when only limited sensor data are available, addressing a key limitation of traditional EKF methods.
- The method is implemented and evaluated in a MATLAB simulation environment, where results demonstrate its effectiveness and robustness in improving localization accuracy under uncertain and incomplete sensor conditions.

The remainder of the paper is organized as follows. Section 2 outlines the problem formulation, detailing the vehicle simulation setup, sensor models, and the integration of neural network-based sensor modeling within the EKF framework. Section 3 presents the numerical simulation results, including vehicle trajectory estimation, neural network training, and the performance evaluation of the EKF with learned sensor models. Finally, Section 4 concludes the paper and highlights the key findings.

2 Problem Formulation

The vehicle dynamics are first simulated along a pre-defined road, generating motion data from two virtual sensors. This data is used to train a neural network to approximate the unknown observation model. The resulting neural model is then integrated with an EKF to estimate the states of vehicle.

2.1 Vehicle Simulation

The simulation model describes the motion of a dynamic system using a state-space representation. The

state vector is defined as:

$$x = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \quad (1)$$

where x_1, x_2 represent the position coordinates and x_3, x_4 represent the corresponding velocity components in north and east directions.

The system is influenced by a control input u , which represents acceleration in a fixed direction given by the angle θ . The state evolves according to the following set of first-order differential equations:

$$\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = x_4 \\ \dot{x}_3 = u \sin \theta \\ \dot{x}_4 = u \cos \theta \end{cases} \quad (2)$$

During the simulation of vehicle, constraints are applied to keep it on a straight road aligned with angle θ just before each time integration step. If the position of vehicle deviates by more than ± 2 meters from the designated road defined by $x_1 = \tan(\theta)x_2$, it will be projected back using

$$\begin{cases} x_1 = x_2 \tan \theta \\ x_2 = \frac{x_2 + x_1 \tan \theta}{(1 + \tan^2 \theta)} \end{cases} \quad (3)$$

Similarly, if the direction of the velocity deviates by more than $\pm 0.2 \text{ m/s}$, it will be corrected by using

$$\begin{cases} x_3 = x_4 \tan \theta \\ x_4 = \frac{x_4 + x_3 \tan \theta}{(1 + \tan^2 \theta)} \end{cases} \quad (4)$$

to maintain the lateral control of the vehicle. A control flag manages bidirectional motion, if the speed exceeds 30 m/s , the flag is set to -1 to reverse acceleration or deceleration, if it drops below 5 m/s , the flag is reset to 1 to resume acceleration. This logic ensures that the vehicle follows the road while alternating the direction of acceleration within a defined speed range.

The numerical integration of the state equations of vehicle dynamics is performed using the classical RK4 method. This method offers high accuracy by computing intermediate slopes at four stages and then combining them to determine the next state. Specifically, RK4 computes weighted increments rk_1 to rk_4 using the system dynamics and combines them as:

$$x_{k+1} = x_k + \frac{1}{6}(rk_1 + 2rk_2 + 2rk_3 + rk_4) \quad (5)$$

2.2 Sensor Models

We have considered two transponders as sensors, placed at fixed coordinates $(0,0)$ and $(173210, 1 \times 10^5)$, as shown in Fig. 1. Transponders are electronic devices used as fixed reference points for positioning

and tracking, particularly in environments where GPS signals are unreliable or unavailable. They emit or reflect signals that enable vehicles to compute relative distances, thereby supporting localization tasks. In this work, the transponders serve as fixed reference points that provide distance-based measurements to estimate the position of a moving vehicle. The sensor measurements are modeled as the squared Euclidean distances between the vehicle position and each transponder, which is a commonly adopted formulation in constrained and nonlinear state estimation frameworks (Simon, 2002). They are given by

$$\begin{aligned} y_1 &= (x_1 - r_{n1})^2 + (x_2 - r_{e1})^2 \\ y_2 &= (x_1 - r_{n2})^2 + (x_2 - r_{e2})^2 \end{aligned} \quad (6)$$

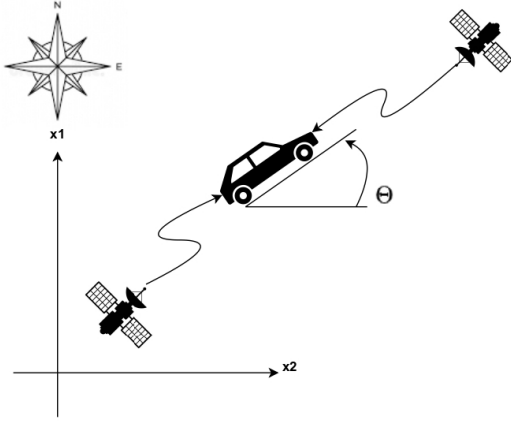


Figure 1: Simulation of a vehicle moving northeast at a 60° angle, tracked using two fixed transponders.

2.3 EKF Algorithm

The Kalman Filter is a recursive algorithm used for estimating the state of a dynamic system from noisy sensor measurements. It is optimal for linear systems with Gaussian noise. However, for nonlinear systems, an EKF is commonly employed. The EKF linearize the nonlinear process and measurement models using first-order Taylor series expansions based on Jacobian matrices, allowing the standard Kalman filtering framework to be applied while accounting for system nonlinearity. In this paper, the EKF is applied to estimate the motion of a vehicle in a 2D plane using simulated sensor data. Consider a nonlinear continuous-discrete dynamic system (Majeed & Kar, 2011).

$$\begin{cases} \dot{x}(t) = f[x(t), u(t), w(t)], & x(0) = x_0 \\ y(t) = h[x(t)] \\ z(k) = y(k) + v(k) \end{cases} \quad (7)$$

Where functions f and h are general nonlinear functions; vector x is the state vector; u is the input vector; y is the output vector, and Δt is sampled output at discrete points with a uniform sampling interval; vectors w and v are process noise and measurement noises and have covariance matrices Q and R , respectively. They are also assumed to be independent between themselves and also with respect to the initial condition $x(0)$, that is,

$$\begin{cases} E(w(k)) = 0; E(v(k)) = 0; E(w(k)v(j)^T) = 0 \\ E(x(0)w(k)^T) = 0; E(x(0)v(k)^T) = 0 \\ E(w(k)w(j)^T) = Q\delta_{kj}; E(v(k)v(j)^T) = R\delta_{kj} \end{cases} \quad (8)$$

As is well known, the algorithm can be divided into two steps: time propagation and measurement correction. These steps are usually referred to as the “prediction” and “correction” phases, respectively. The initial state estimate $\hat{x}(0)$ and the corresponding covariance matrix $P(0)$ are assumed initially and then propagated through the prediction step of the EKF. The complete computational procedure of the EKF algorithm is summarized below (Majeed & Kar, 2010).

Prediction (time update):

$$\begin{cases} \tilde{x}(k+1) = \hat{x}(k) + \int_{t_k}^{t_{k+1}} f[x(t), u(t)] dt \\ \tilde{P}(k+1) = \Phi(k+1, k)\hat{P}(k)\Phi^T(k+1, k) + \dots \\ \Psi(k+1, k)B(k)Q(k)B^T(k)\Psi^T(k+1, k) \end{cases} \quad (9)$$

$$\Phi(k+1) = I + A(k)\Delta t + A^2(k)\frac{\Delta t^2}{2!} + \dots$$

$$\Psi(k+1) = I\Delta t + A(k)\frac{\Delta t^2}{2!} + A^2(k)\frac{\Delta t^3}{3!} + \dots$$

$$A(k) = \left. \frac{\partial f[x(t), u(t)]}{\partial x} \right|_{x=\hat{x}(k)}, B(k) = \left. \frac{\partial f[x(t), u(t)]}{\partial u} \right|_{x=\hat{x}(k)}$$

Correction (measurement update):

$$\begin{cases} \tilde{y}(k) = h[\tilde{x}(k)] \\ K(k) = \tilde{P}(k)C^T(k)[C(k)\tilde{P}(k)C^T(k) + R(k)]^{-1} \\ \hat{x}(k) = \tilde{x}(k) + K(k)[z(k) - \tilde{y}(k)] \\ \hat{P}(k) = [I - K(k)C(k)]\tilde{P}(k)[I - K(k)C(k)]^T + \dots \\ + K(k)R(k)K^T(k) \end{cases} \quad (10)$$

Where

$$C(k) = \left. \frac{\partial h[x(t)]}{\partial x} \right|_{x=\hat{x}(k)}$$

2.4 Neural Network Modelling

In this section, we develop and apply a FFNN to build a nonlinear model of sensors based on input-

output data. We used FFNNs with two hidden layers to describe vehicle measurements z_1 and z_2 , which correspond to squared range measurements from two transponders, as used in the EKF formulation.

Training a NN for a sensor model involves building a model that can predict sensor outputs, calibrate noisy readings, or infer high-level information from raw sensor data. The input vector to the neural network is composed of the state variables of vehicle dynamics represent the north and east positions and their squares. The input vector is given by

$$x = [x_1 \ x_2 \ x_1^2 \ x_2^2] \quad (11)$$

The output of neural network is either z_1 or z_2 , corresponding to the nonlinear range measurement from each transponders. The NN is structured as a feed-forward multi-layer perceptron with the following configuration:

- Input layer: 4 nodes (for x_1, x_2, x_1^2, x_2^2)
- Hidden Layer 1: N_{HL1} neurons with hyperbolic tangent activation
- Hidden Layer 2: N_{HL2} neurons with hyperbolic tangent activation
- Output Layer: 1 neuron with linear activation

The activation function used in the hidden layers is the hyperbolic tangent function, given by:

$$f(y) = \frac{1 - e^{-\lambda y}}{1 + e^{-\lambda y}}, \quad f'(y) = \frac{2\lambda e^{-\lambda y}}{(1 + e^{-\lambda y})^2} \quad (12)$$

with slope parameter $\lambda = 0.25$ applied uniformly across all layers. Before training the neural network, the input and output data are preprocessed to ensure consistent convergence during optimization. Each input variable consisting of x_1, x_2, x_1^2, x_2^2 , as well as the sensor output (either z_1 or z_2), is normalized or scaled into the range $[-0.9, 0.9]$. This normalization ensures stable training behavior and allows the network output to be interpreted in their original physical units (Mohamed & Dongare, 2018). The NN is trained using the standard backpropagation algorithm, where the cost function minimized during training is the mean squared error (MSE) between the predicted outputs and the actual outputs:

$$E = \frac{1}{N} \sum_{i=1}^N \left(y_{\text{pred}}^{(i)} - y_{\text{true}}^{(i)} \right)^2 \quad (13)$$

Weight updates in the network are governed by the backpropagation rule with momentum Ω :

$$\Delta W_{ij}^{(t+1)} = -\mu \cdot \frac{\partial E}{\partial W_{ij}} + \Omega \cdot \Delta W_{ij}^{(t)} \quad (14)$$

Where W_{ij} is the weight from node j to node i and $\Delta W_{ij}^{(t)}$ is the weight change from the previous iteration

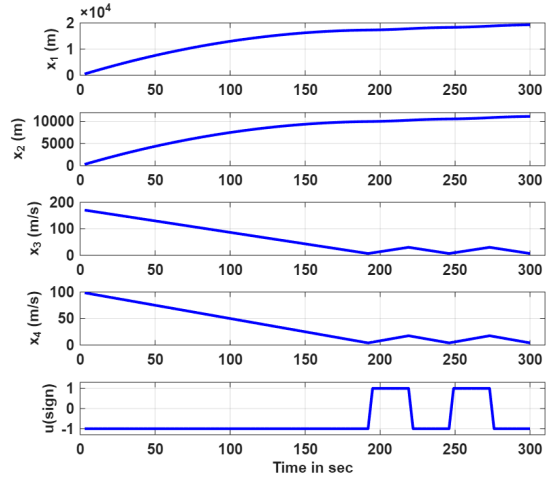


Figure 2: Vehicle Position and Velocity in north and eastern directions

3 Numerical Simulation Results and Discussions

3.1 Vehicle Simulation Results

The vehicle simulation models motion using the RK4 method, starting from the initial state $x(0) = [0; 0; 173.2; 100]$, where the first two elements represent the vehicle's initial position in the north and east directions, and the last two represent the initial velocities. The system evolves under a discrete control input $u(t) = [-1, 0, 1]$, which affects the vehicle's acceleration and steering behavior. The RK4 method is used to compute state updates at each time step, ensuring accurate numerical integration of the nonlinear dynamics. The simulation results are summarized in Figure-2 and illustrates the vehicle's trajectory through its position and their velocities in the north and east directions.

3.2 Vehicle Estimation Results

The EKF provides accurate and robust state estimation for position and velocity in the north and east directions, as shown in Figures 3 to 5. Although it initially diverges due to uncertainty in initialization, it quickly converges as measurements are assimilated. The estimated position closely tracks the true trajectory while effectively filtering noise. Velocity estimates, although not directly measured, align well with actual values, demonstrating the ability of the EKF to infer hidden states and maintain reliable performance under noisy conditions.

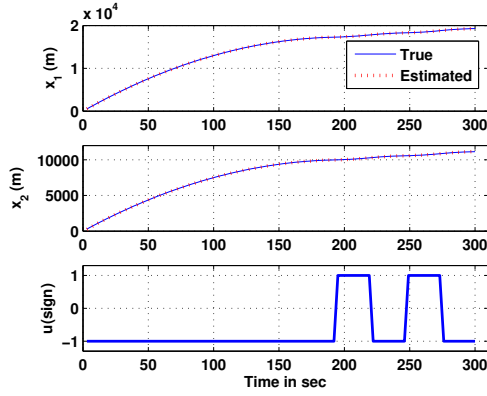


Figure 3: Estimated position tracks the true position of Vehicle

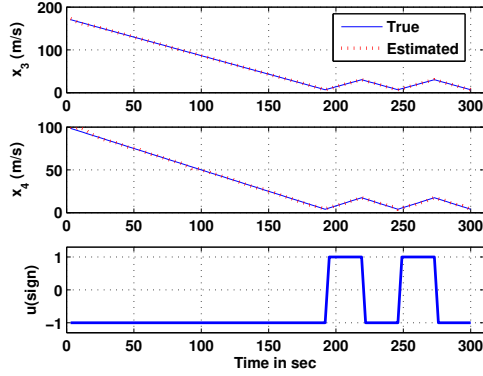


Figure 4: Estimated Velocity of Vehicle tracks the true Velocity

3.3 Neural Modeling of Sensors

The performance of the neural networks was evaluated by comparing the sensor measurements (z_1 and z_2) with the corresponding neural network outputs over time, as illustrated in Figure-6 and 7. These plots highlight the networks' ability to accurately ap-

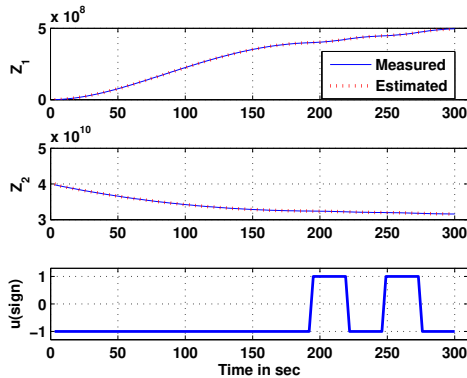


Figure 5: Time history of sensor measured and estimated responses

proximate the nonlinear measurement functions employed in the Kalman Filter framework. Separate NNs were trained for each measurement, representing the squared distances from the vehicle to individual transponders. Training was performed using backpropagation for over 7000 iterations, with each network comprising two hidden layers. The results indicate that both networks achieved strong convergence and effectively captured the underlying nonlinear relationships.

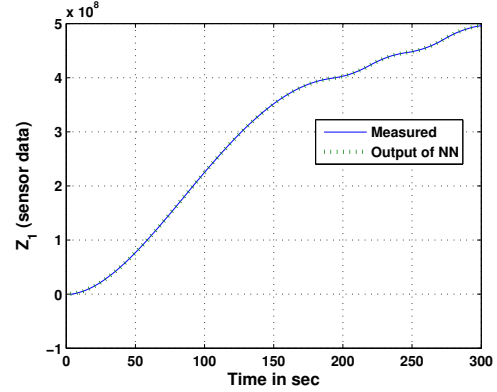


Figure 6: The NN predicts the sensor measurement z_1

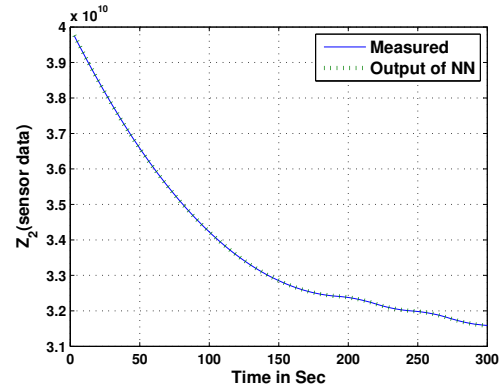


Figure 7: The NN predicts the sensor measurement z_2

4 Integration of EKF and Neural Model

In advanced vehicle localization, integrating neural networks with the EKF enhances accuracy and robustness by effectively handling nonlinearities and modeling uncertainties. To evaluate the robustness and efficacy of the EKF integrated with neural models, two scenarios were examined, focusing on sensor modeling limitations and data unavailability.

4.1 EKF with a limited sensor model

In the considered localization framework, transponders act as reference points, providing squared range measurements that are nonlinear functions of the vehicle's north and east positions. The measurement model at coordinate (r_{n2}, r_{e2}) is given by

$$z_2 = (x_1 - r_{n2})^2 + (x_2 - r_{e2})^2 \quad (15)$$

We assumed that measurement model for z_1 is unknown and a neural model was established using a FFNN trained via the backpropagation algorithm based on the available data of z_1 . The established measurement model contains two hidden layers with 12 neurons in first layer and 23 neurons in second layer and is given by

$$z_1 = F(x, w1, w2, w3) \quad (16)$$

The NN consists of three layers of weights. The first weight matrix $w1$ has dimensions 12×5 , indicating that the first hidden layer has 12 neurons and the input layer has 5 inputs (including bias). The second weight matrix $w2$ has dimensions 23×13 , meaning the second hidden layer contains 23 neurons and receives 13 inputs (which may include the 12 outputs from the previous layer plus a bias). Finally, the output layer weight matrix $w3$ has dimensions 1×24 , indicating a single output neuron that receives 24 inputs from the previous layer (likely the 23 outputs from the second hidden layer plus bias). The match between the data of z_1 and neural model is given in the Figure 6.

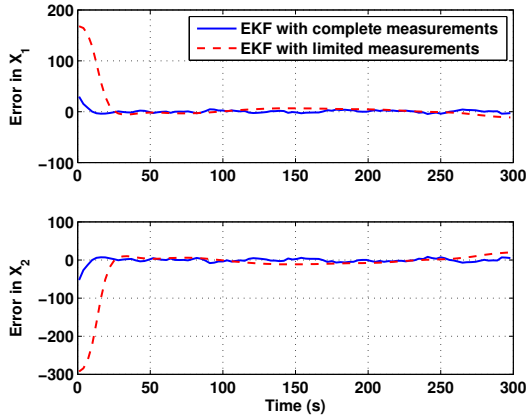


Figure 8: The position estimation error due to limited measurements

The state estimation results are analyzed using a neural model represented by (16) to correct the unknown measurement model. The estimated states

closely match the true trajectories of the vehicle's position. The position estimation errors are compared with those obtained using the EKF with complete measurement models (6), as shown in Figure 8. The initial mismatch observed in the figure is due to a delay in the establishment of the neural model. In the estimation of velocity states, although an initial mismatch is observed, the estimates rapidly converge to the true trajectory, as illustrated in Figure 9. This demonstrates the effectiveness of the proposed approach within the accepted range of errors, even under sparse observations.

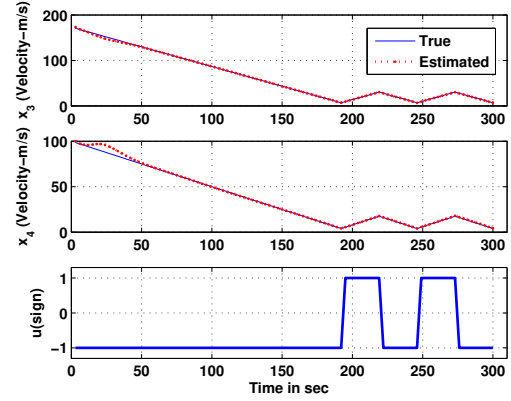


Figure 9: The velocity estimates with limited measurements

4.2 EKF with a limited sensor data

In this case, the vehicle is assumed to operate under conditions where the measurement z_1 is unavailable, such as during tunnel traversal or GPS-denied scenarios. To address this, the previously developed neural model for z_1 is used to generate synthetic sensor measurements from vehicle dynamics inputs and is incorporated into the estimation framework as the measurement equation. Figure 9 presents the position estimation error under the absence of z_1 measurements, while the corresponding mean square errors of the estimated states are summarized in Table 1. The results indicate an increase in estimation error due to reduced sensor availability, primarily caused by the neural model's approximation error, which lies in the range of 10%–20%.

Future work will focus on improving model accuracy through more diverse training data and advanced architectures such as radial basis function (RBF) neural networks. In addition, real-time implementation on embedded platforms will be investigated to validate the proposed approach under practical autonomous navigation scenarios.

Table 1: Mean square error in state estimation.

State Estimates	MSE in EKF Estimation		
	Ideal	Limited sensor models	Limited sensor data
N-position	16.2117 ^a	50.4748	75.8845
E-position	32.8214	73.5628	115.7428
N-velocity	0.0832 ^b	0.3008	13.8628
E-velocity	0.2380	0.7527	15.6421

^aValues in meters, ^bvalues in meters per second.

5 CONCLUSIONS

This work presented a hybrid approach for vehicle position estimation in environments with sparse or unreliable sensor data by integrating neural network-based sensor models with the EKF. A Feed Forward Neural Network was trained offline to learn complex, nonlinear sensor behaviors, effectively replacing traditional measurement models that are difficult to define analytically. These learned models were then incorporated into the EKF framework to estimate vehicle states under limited observation conditions, such as GPS outages. Simulation results demonstrated that the NN-enhanced EKF maintained robust and accurate performance, even with partial sensor information. The findings highlight the viability of this approach for real-world applications like autonomous driving and advanced driver-assistance systems, where resilience to sensor degradation is critical.

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